

Perform unit testing on basic functions (add, subtract, multiply, divide) in C++

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Subject: MDM (Software Automated System)]

*math_functions.cpp

```
int add(int a, int b) {  
    return a + b;  
}  
int subtract(int a, int b) {  
    return a - b;  
}  
int multiply(int a, int b) {  
    return a * b;  
}  
double divide(int a, int b) {  
    if (b == 0) throw std::invalid_argument("Division by zero");  
    return static_cast<double>(a) / b;  
}
```

*test_math_functions.cpp

```
#include "gtest/gtest.h"  
#include "math_functions.cpp"  
TEST(MathTest, Addition) {  
    EXPECT_EQ(add(3, 2), 5);  
    EXPECT_EQ(add(-1, -2), -3);  
}  
  
TEST(MathTest, Subtraction) {  
    EXPECT_EQ(subtract(10, 4), 6);  
    EXPECT_EQ(subtract(5, 7), -2);  
}  
  
TEST(MathTest, Multiplication) {  
    EXPECT_EQ(multiply(3, 4), 12);  
    EXPECT_EQ(multiply(-2, 5), -10);  
}  
  
TEST(MathTest, Division) {  
    EXPECT_DOUBLE_EQ(divide(10, 2), 5.0);  
    EXPECT_THROW(divide(5, 0), std::invalid_argument);  
}
```